DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Supplementary Examination Summer -2024

Course: B. Tech **Semester : VII**

Subject Code & Name: BTETOE704F E Waste Management

Branch : Electronics and Communication Engineering / Electronics and Telecommunication Engg.

Max Marks: 60 **Date: 08/07/2024** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	What is E waste & explain common types of electronic devices that contribute to e-waste?	**CO 1**	**6**
B)	Explain the rules for E waste prescribed by Govt. of India	**CO 2**	**6**
C)	Describe the hazardous components present in electronic waste.	**CO 1**	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	Explain the concept of the circular economy in the context of e-waste management.	**CO 1**	**6**
B)	Explain the concept of "Reduce, Reuse, Recycle"	**CO 2**	**6**
C)	Describe the process and importance of e-waste health risk assessment.	**CO 2**	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	What are the challenges associated with the recovery of materials from e-waste?	**CO 1**	**6**
B)	Describe the steps involved in the metal recovery process from electronic waste.	**CO 2**	**6**
C)	What are the best practices for e-waste management?	**CO 1**	**6**
Q.4	**Solve Any Two of the following.**		**12**
A)	What is Life Cycle Assessment (LCA) and how is it applied to electronics?	**CO2**	**6**
B)	Discuss the technological innovations used in the recycling of electronic waste.	**CO 2**	**6**
C)	Explain the role of governmental and non-governmental organizations in addressing environmental and public health concerns.	**CO 1**	**6**
Q. 5	**Solve Any Two of the following.**		**12**
A)	Discuss benefits of using LCA in the design and development of electronics.	**CO 2**	**6**
B)	Explore innovative e-waste management strategies that prioritize sustainability and environmental protection.	**CO 2**	**6**
C)	Explain the environmental impact of improper metal disposal from electronic waste.	**CO 1**	**6**

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